

Epibiont Monitoring on Shellfish Aquaculture Gear

Protocol ID	dealteris_2004_hb111
Sector	Molluscs & echinoderms (native oyster)
Usability tier	Tier 3 — Partnership (collection farmer-runnable; ID & analysis partner-supported)
What it monitors	The abundance, diversity, and cover of the epibiont community — the sessile invertebrates and associated fauna living on the shellfish aquaculture gear.
Source paper	DeAlteris, J.T., Kilpatrick, B.D. & Rheault, R.B. (2004). A comparative evaluation of the habitat value of shellfish aquaculture gear, submerged aquatic vegetation and a non-vegetated seabed. <i>Journal of Shellfish Research</i> , 23(3), 867–874.
Related	This survey also records the mobile fauna and can compare the gear against nearby habitats; those indicators (mobile fauna, community composition) are covered separately by ferriss_2021_hb121 / _hb211.
Associated tools	Part of a bundle: this normalized protocol, a field datasheet, and a stats workbook. Download links are added once the bundle is hosted.

Source: https://digitalcommons.uri.edu/favs_facpubs/26

Adapted into a general, gear-agnostic method for AquaPIP. Open-access (peer-reviewed, Journal of Shellfish Research; CC BY-NC-SA via DigitalCommons@URI).

Introduction

Shellfish aquaculture gear — the cages, bags, and shells of an oyster grow-out system — is itself a habitat: over a season it accumulates a community of attached and associated organisms. This protocol turns that into a repeatable measure of how much life the gear hosts, which informs the farm's ecological role.

This paper compared the fauna associated with oyster rack-and-bag gear against eelgrass and bare seabed over a year in a Rhode Island estuary. The authors highlight that the physical structure and surface area of the gear support a dense community of sessile invertebrates and mobile fauna — comparable to, and in abundance exceeding, submerged vegetation — so structure and surface area are measured alongside the organisms.

Parameters measured

This protocol targets a single indicator — the epibiont community on the gear. Monitoring linked to it (green) is the epibiont survey; context monitoring (grey) captures the gear's structure and the environment so the community can be interpreted and stood behind (abundance scales with surface area). Each parameter is listed with its unit and a working definition.

■ monitoring linked to a selected indicator ■ context monitoring

Epibiont community on the gear

- **Sessile-epibiont identity & cover (% area)** — the attached organisms on the cage, bags, and shells — sponges, hydroids, bryozoans, ascidians, algae — by taxonomic group (phylum/class), and the percentage of surface each covers, from fixed quadrats.
- **Associated fauna abundance (count)** — the mobile animals living on/among the gear (small fish, crustaceans, molluscs > ~5 mm), from an enclosure sample.
- **Taxa richness & diversity** — number of taxa and a diversity index (e.g. Shannon) of the associated community.

Gear structure — context (abundance scales with it)

- **Emergent surface area (cm² per m² of seabed)** — the surface area the gear presents (cage + bags + shells) per unit of seabed — the habitat structure the community grows on.
- **Oyster / gear metrics** — oyster size (length, width) and gear type/configuration.

Environment — context monitoring

- **Water temperature (°C), salinity (ppt), dissolved oxygen (mg L⁻¹)** — site conditions at each seasonal sample.
- **Sediment grain size** — seabed character beneath the gear, for context.

Requirements

- **Personnel [estimate]:** a small boat crew to set and recover the enclosure sampler on the gear (farmer-runnable); and a partner able to identify the sessile and mobile fauna and run the diversity analysis.
- **Time per visit [estimate]:** a seasonal sampling session (set the enclosure, recover it and a few bags, run quadrats); most of the effort is subsequent lab identification.

Sampling frequency & timing

- **Framework guidance.** The MEL framework sets the mollusc native-epibiota indicator (1.1.1) at seasonal, sampling “at intervals sufficient to build an accurate depiction of the process over time.”
- **What the paper reports.** This paper samples once per season across a full year (four seasons), with three replicate gear units per season.
- **Recommended to detect an effect.** Sample at least once per season across a full year (colonisation and fouling change with season), and repeat across [2–3+] years so year-to-year variability can be separated from a farm signal.

Materials

Materials are roles to fill, not products to buy. Collection is straightforward; the identification is the specialist part.

► Gear varies site to site — use your best judgement

The source used a lift-net enclosure set by a diver, fixed quadrats, and specific stats software. Substitute an enclosure net, quadrat frame, and stats package you have; in shallow water the enclosure can be set from the surface. The specialist need is taxonomic identification. Nothing below is mandatory as a specific product.

- **For the survey:** an enclosure sampler sized to the gear (e.g. a fine-mesh [2 mm] lift-net set under and recovered around a cage); a fixed quadrat [e.g. ~0.02 m²] for sessile % cover on the cage/bags; calipers / measuring board; a balance; a microscope and taxonomic ID references; and a stats package (species richness, Shannon-Wiener diversity, ANOVA).

Methods

Single-method protocol — enclose and sample one gear unit, quantify the sessile cover on it, and identify the associated community. Collection is farmer-runnable; identification and analysis are partner-supported (see tier).

Enclosure sampling of the gear

↳ *Linked indicator: supports an abundance and diversity of native epibiota — the community on the aquaculture gear — match: Partial. Full mapping in Appendix A.*

Materials. a fine-mesh enclosure/lift-net sized to a gear unit; a fixed quadrat; calipers/measuring board; a balance; sample bags; microscope + ID references; stats package.

- **Preparation.** Fix the sampled unit — a set number of gear units [e.g. 3] per visit, cleaned a consistent time before sampling [e.g. 4–6 weeks] so fouling is representative. Set the enclosure net under a cage a short time ahead [e.g. ~2 weeks] so disturbed mobile animals return.
- **Collection.** Recover the enclosure so it fully encloses the gear unit; collect all free-swimming fauna [$> \sim 5$ mm] plus a few [e.g. 3] bags. At several fixed points [e.g. 5] on the cage and bags, record sessile-epibiont percentage cover by taxonomic group using the quadrat.
- **Measurement.** Estimate the emergent surface area of the gear unit (cage + bags + a sample of shells) and record oyster size — abundance scales with surface area. Identify and count the associated fauna to the finest practical level; weigh where useful.
- **Analysis.** Per gear unit, summarise sessile % cover by group, associated-fauna abundance, taxa richness, and a diversity index, expressed relative to surface area / per m^2 of seabed. Compare across seasons. A simple summary answers the indicator; the source's ANOVA / diversity-index comparison is the partner-supported statistical step.

Expect these failure modes and constraints (and manage them)

Each constraint is grouped under the method or stage it affects, with how to manage it.

Enclosure sampling

- **Mobile-fauna escape on recovery** — fast animals can leave as the enclosure is lifted; set the net ahead of time and recover it fully around the gear to limit loss; treat mobile counts as a relative index.
- **Taxonomic resolution** — sessile groups are practically resolved to phylum/class and mobile fauna to species; fix the level up front and keep it constant so results stay comparable.
- **Cleaning disturbs the community** — routine gear cleaning removes fouling; sample a consistent interval after cleaning so the fouling stage is comparable between visits.

Across design & interpretation

- **Confounding by surface area** — more gear surface hosts more organisms regardless of any farm effect; always express abundance/richness relative to emergent surface area.
- **Single site / season** — one site over one year can be swamped by year-to-year variation; repeat across years before drawing firm conclusions.

Data handling

Use a consistent labelling convention: site — gear unit — season — date. Enter cover, counts, weights, and surface-area measurements into the companion workbook; its data-entry columns mirror the datasheet exactly. Keep reference images of the common sessile groups for consistent identification. [Insert your own site photo] where useful.

Methods referenced from other sources

Two steps follow established methods. Resolve and verify a stable link for each before relying on it.

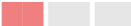
- **Enclosure sampling of small fishes & decapods** — the enclosure/lift-net approach in the Collection step. Rozas, L.P. & Minello, T.J. (1997). Estimating densities of small fishes and decapod crustaceans in shallow estuarine habitats: a review of sampling design with focus on gear selection. *Estuaries*, 20, 199–213. [[resolve stable link](#) — [verify](#)]
- **Species richness / diversity indices** — the richness, Shannon-Wiener diversity, and evenness calculations in the Analysis step. (Standard ecological-diversity indices; the source used Ecological Methodology software.) [[resolve stable link](#) — [verify](#)]

Appendix A — MEL objective table

Reference: this protocol targets one indicator; the row below is the full form of the per-method “Linked indicator” line above. Framework codes appear only here.

Goal	Monitoring method	MEL Objective	Metric(s)	Match
GOAL 1 Molluscs and echinoderms are farmed in a way that supports an abundance and diversity of associated native species.	Enclosure sampling + sessile-cover quadrats on the gear	Objective 1.1 (Ind. 1.1.1) — Farming supports an abundance and diversity of native epibiota (organisms on the farmed biomass)	Abundance & diversity of epibiota — sessile % cover by group, associated-fauna abundance, taxa richness and diversity, relative to surface area	 Partial

Match scale. How closely the method's metric matches the objective's.

-  **Exact** — measures the objective's metric as specified.
 **Partial** — measures a proxy for, or subset of, the metric, or needs an added condition such as a control site.
 **Indirect** — gives relevant evidence toward the objective (a native species present/active) but not the specified metric.

Not mapped (context). Gear structure (emergent surface area, oyster size, gear type) is recorded because abundance scales with it, and water temperature, salinity, dissolved oxygen, and sediment are recorded as environmental context — none are themselves mapped. Mobile fauna and the gear-vs-vegetation-vs-bare-seabed comparison are the subjects of separate indicators (covered by ferriss_2021_hb121 / _hb211).

Appendix B — Provenance & practitioner decisions

- **Adapted from:** DeAlteris, J.T., Kilpatrick, B.D. & Rheault, R.B. (2004). A comparative evaluation of the habitat value of shellfish aquaculture gear, submerged aquatic vegetation and a non-vegetated seabed. *Journal of Shellfish Research*, 23(3), 867–874. https://digitalcommons.uri.edu/favs_facpubs/26 (peer-reviewed; open access, CC BY-NC-SA).
- **Indicator set:** This protocol targets a single indicator — native epibiota on the aquaculture gear (match Partial; sessile taxa resolved to phylum/class). Tier T3: collection is farmer-runnable, but identification and the diversity analysis need partner support. Fills the mollusc epibiota gap left by removing the non-peer-reviewed hossain_2013. The same fieldwork also measures mobile fauna and a habitat comparison, but those indicators are covered by ferriss_2021 and are out of scope for this single-indicator protocol.
- **Decisions left to the practitioner [bracketed]:** these are choices this protocol leaves open for the site — not gaps the authors reported. Set them locally: number of gear units sampled and the cleaning interval before sampling; the enclosure and quadrat sizes; the practical taxonomic level; how surface area is estimated; the instruments for water quality; and whether the optional statistical comparison is run.